Matthew Schwartz

mzschwartz5@gmail.com
(610) 329-4222

Work Experience

Cesium, 3D Graphics Engineer *June 2025 - current*
• Build a WebGL-based rendering engine for viewing high quality 3D geospatial data.
• Improve user experience with massive performance upgrades to terrain picking and traversal flows.
• Prototype rapid solutions and troubleshoot for customers via forums and GitHub issue triage.

Amazon Web Services, Software Development Engineer *April 2022 - August 2024*
• Developed Tying, a Java-based engine for automated patching and OS migration of Amazon hosts.
• Designed and authored integration and end-to-end load tests for the Tying engine.
• Improved dev-ops tooling and monitor service metrics proactively and in on-call capacity.
• Expanded and maintain CI/CD and distributed cloud computing infrastructure-as-code.

Epic Systems, Software Developer *August 2019 - August 2021*
• Developed and design the App Orchard website, a marketplace for third-party healthcare apps.
• Implemented RESTful web APIs to expose health record data according to the HL7 FHIR standard.
• Led transition of App Orchard site from ASP.Net framework to single-page React app.

Education

University of Pennsylvania, 2025 M.S. in Computer Graphics and Game Technology, 3.93 GPA

Rice University, 2019 B.S. in Computational Physics; Minor in Applied Mathematics, 3.88 GPA

Projects

***Cubit* (Maya Plugin),** https://github.com/mzschwartz5/cubit
• Soft-body deformation and destruction in a single plugin using PBD with voxel gram-schmidt constraints, and custom tools for painting simulation mesh constraints and live interaction.

Fluid Surface Mesh Shader, https://github.com/mzschwartz5/Breakpoint
• A mesh-shading based fluid surface reconstruction demonstration with a 4x performance improvement compared to the original (https://dl.acm.org/doi/abs/10.1145/3651285).

CUDA Pathtracer, https://github.com/mzschwartz5/CUDA-Path-Tracer
• A pathtracer built on CUDA with support for various PBR materials, GLTF model loading, denoising, and various performance optimizations.

Node Based SDF Modeling Tool, https://github.com/mzschwartz5/Node-Based-SDF-Modeling-Tool
• An OpenGL node-graph tool for quickly creating procedural 3D models, via raymarched SDFs.

Mini Minecraft, https://github.com/mzschwartz5/Mini-minecraft
• An OpenGL Minecraft recreation with support for procedural terrain with multithreaded loading, dynamic shadow mapping, atmosphere and post process shaders, physics, and much more.

Technologies: C++, DirectX, CUDA, HLSL, OpenGL, QT, Unreal Engine, WebGPU, Unity, Maya